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Cryptocurrencies and Bubble Wealth

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Abstract

Americans now hold over \$3.7 trillion in cryptocurrencies. Has \$3.7 trillion in wealth been created? From the standpoint of economic theory, the answer is no. The wealth of a society consists of its real assets that produce consumable goods and services. Unless a cryptocurrency provides some type of convenience yield how could it create wealth? On the other hand, everyone who holds the currency thinks of it as wealth because it can be sold and converted into consumption. This short note takes a step in resolving the apparent paradox by presenting a very simple numerical example of the creation of what I call *Bubble Wealth*.

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The example begins with an “economy” that consists of three individuals: A, B, and C. At the outset in period 0, the wealth of each of them consists of 10 units of a “stock” that pays out in terms of future consumption of goods and services. Consequently, both the aggregate social wealth, the claims to future consumption, and the perceived social wealth, calculated by summing the perceived wealth of all the individuals, are both 30 units as shown in the exhibit below.

In period 1, individual A invents a digital coin denoted by Coin. This Coin provides the owner with no rights to future consumption and pays no interest, but it is designed to trade freely. At this point, there is still no difference between actual and perceived social wealth.

In period 2, A sells Coin to B for one unit of stock. After the trade closes, A owns 11 units of stock, B owns 9 units plus Coin, and C still has 10 units. The aggregate social wealth measured in terms of rights to future consumption of goods and services remains 30 units, but the perceived wealth has risen to 31 because B believes Coin is worth at least the 1 unit that he paid for it. The results are shown in the table below.

In period 3, C buys Coin from B for 2 units based on the belief that its price will rise in the future. As shown in the exhibit the holdings A, B, and C are now: 11, 11, and 8 + Coin. The

perceived wealth has increased to 32 because C concludes that Coin is worth at least the 2 units that he paid for it.

In period 4, the investors recognize that Coin has no value in terms of future consumption and the price collapses. The perceived social wealth drops back to the actual social wealth of 30. The net effect of the series of transactions is the transfer of 2 units of stock from C, who was left holding the bag when the collapse occurred, to A and B, who each gain 1 unit.

Bubble Wealth Numerical Example

Period	A holdings	B holdings	C holdings	Social Wealth	Perceived Social Wealth
0	10	10	10	30	30
1	10+Coin	10	10	30	30
2	11	9+Coin	10	30	31
3	11	11	8+Coin	30	32
4	11	11	8	30	30

The example could be extended by adding more individuals and more trading periods, but the basic point remains unchanged. As long as Coin is trading, it creates *bubble wealth* defined as the difference between actual social wealth related to future consumption of goods and services and perceived social wealth. This gap between the perceived and actual social wealth persists until there is a crash and the value of Coin drops to its fundamental value of zero. At that point, it becomes clear that the invention of Coin did not create wealth, but only promoted its transfer – in the case of the example from C to A and B.

One problem with the simple example is that there is no reason for trading in Coin to start in the first place. In the real world, however, there may be a number of catalysts. First, trading in Coin is an ideal speculative vehicle for gambling. As Warren Buffett observes the gambling instinct is very strong and crypto trading provides an ideal outlet. Second, trading Coin could be

a way for those wishing to avoid government oversight to move funds around to avoid taxes or engage in more nefarious activities. Third, in the example Coin provides absolutely no services, but in reality Coin could be a substitute means of exchange thereby providing a convenience yield. To date, however, there is little evidence that cryptocurrencies are being used as a means of exchange to any meaningful degree. Whatever the reason that trading starts, if the price begins to rise then in a world of incomplete information and heterogeneous beliefs, the price increase could become self-fulfilling in a manner such as that originally described by Harrison and Kreps (1978). People want to get in on the game because they see other people making speculative profits.

The fact that Coin does not create social wealth does not mean its crash cannot destroy wealth in a more realistic context than the simple example. If market frictions and politics are involved, wealth transfers caused by the crash of Coin could have important real effects. Furthermore, if individuals make spending decisions based on their perceived wealth there could be macroeconomic effects.

Finally, just as initial trading in Coin requires a catalyst, so does the crash. If investors currently holding Coin want to convert their holdings into consumption of goods and services, they have to sell. Because Coin provides no rights to future consumption, there are no fundamental investors who would be willing to buy Coin. The only buyers are other speculators. If the price has risen to the point where no speculator is willing to pay more than the last transaction price, the seller will have to accept a discount. As soon as the price falls, the speculative reason for holding Coin evaporates and the price collapses.

REFERENCES

Harrison, Michael J., and David M. Kreps, 1978, Speculative investor behavior in a stock market with heterogenous beliefs, *Quarterly Journal of Economics*, 92 (2), 323-336.